

Predict, Optimize, Deliver: Demand Forecasting and Resource Optimization for a Market Research Firm

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Abstract

In professional service industries, where human resources could drive both cost and value, accurate forecasting and intelligent resource allocation were critical to sustaining operational performance. To this end, this paper developed an integrated forecasting and optimization framework to address demand uncertainty and resource allocation challenges in a market research firm. The study compared ARIMA, Holt-Winter's Exponential Smoothing, Prophet, and LSTM models in terms of service level accuracy, assessed via MAE, RMSE, and MAPE across multiple training-test splits and a 30-day hold-out period. Among the evaluated methods, Prophet demonstrated the highest overall effectiveness, offering a robust balance between forecast accuracy, interpretability, and operational usability, especially for short-term planning horizons. The selected forecasts were then operationalized using a Pyomo-based prescriptive optimization model that allocated workforce resources in alignment with anticipated service demand, subject to constraints such as employee availability and client-specific requirements. Forecast bands derived from RMSE were introduced to simulate over- and under-forecasting scenarios, strengthening the model's resilience and responsiveness in allocation strategies. The final framework was deployed within an interactive Power BI environment, providing dynamic visualizations and supporting agile, cost-effective workforce planning. Overall, this study contributed a practical, data-driven approach that bridged predictive analytics and operational planning, offering actionable insights for resource optimization in professional service industries.

CCS Concepts

• Information Systems; • Decision Support Systems; • Data Analytics;

Keywords

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1 INTRODUCTION

For the subject company, where client needs are highly variable, effective resource allocation is essential to maintaining operational efficiency and profitability. Consulting companies experience fluctuating service requests tied to client behavior and project timelines [1]. Since manpower is the primary asset in such firms, delays or inefficiencies in staffing lead to missed deadlines and potential revenue loss [2]. According to Resource-Based Theory (RBT), optimal internal resource deployment contributes to sustained competitive advantage [3], especially in outsourced delivery models such as the studied company, which conducts 70% of its operations through its Manila-based service center.

In this study, the subject firm handles client-specific “demand” defined in hours across service categories, each varying in complexity and processing time. Demand surges unpredictably, often requiring overlapping workforce deployment. For example, Client 1 typically requires 2-3 full-time equivalents (FTEs), increasing to 4 during peak periods, while Client 2 presents more erratic spikes. Manual workforce planning like MS Excel fails to adapt to such volatility. While large enterprises leverage ERP platforms like SAP or Oracle [4], smaller firms must choose between prebuilt tools and custom models [5]. Forecast accuracy is critical as failures like those experienced at Heathrow Airport have shown that poor demand prediction can trigger systemic disruption [6]. Selecting the right forecasting approach requires evaluating trade-offs between accuracy, complexity, and interpretability, especially in dynamic service environments [7].

This study proposed an integrated solution using a Predict-Then-Optimize (PO) framework that combined time-series forecasting with network flow-based resource allocation. Demand was first forecasted using four techniques, specifically ARIMA, Holt-Winters, LSTM, and Prophet. These were used to inform an optimization

model that respected real-world constraints such as work-hour caps and shared staffing. The results were visualized in a Power BI dashboard for operational transparency. The objective was to enhance resource efficiency, align staffing to fluctuating service demand, and demonstrate how advanced analytics can empower midsize firms. This work contributed by identifying the most suitable forecasting method for a professional servicing firm, operationalizing data-driven allocation strategies, and delivering actionable insights through a usable, visual platform. This supported SDG 8 by promoting sustainable employment practices, SDG 9 by advancing data-driven decision-making in mid-sized firms, and SDG 17 by fostering collaboration between analytics teams, operations, the academe and strategic planners.

2 LITERATURE REVIEW

Time series forecasting assumes that future outcomes are extensions of historical data, governed by trends, seasonality, cycles, and randomness [8]. For industries that rely on time-bound outputs, particularly the sample company where staffing and demand shift rapidly, accurate forecasting becomes crucial. To this end, this study investigates four prominent forecasting methods, specifically ARIMA, Holt-Winters, LSTM, and Prophet, while assessing their performance using RMSE, MAE, and MAPE metrics.

We begin with ARIMA, a classical yet widely adopted model. Its strength is observed in handling stationary time series, with well-documented applications in supply chain [9], healthcare [10], and cloud resource planning [11]. However, as demand signals become more volatile and irregular in modern service operations, ARIMA's limitations in capturing seasonality and nonlinear trends emerge as critical gaps, thereby motivating the search for more adaptive alternatives.

In response, Holt-Winters Exponential Smoothing builds on ARIMA's strengths by integrating level, trend, and seasonal components through flexible smoothing parameters (α, β, γ) [12, 13]. This allows the model to dynamically adjust to recent observations, making it well-suited for data that evolve over time. Depending on whether seasonal variations remain constant or scale with magnitude, the model shifts between additive and multiplicative forms [14]. Such adaptability has proven useful in real-time applications, including epidemiological forecasting [14].

Yet, even with enhanced seasonality handling, both ARIMA and Holt-Winters assume largely linear patterns. For service environments characterized by erratic shifts and nonlinear dependencies, Long Short-Term Memory (LSTM) networks offer a more powerful alternative. LSTM models, equipped with memory cells and gating mechanisms, are capable of learning both short- and long-term dependencies in time series data [15, 16]. As recent studies show, LSTM has consistently outperformed ARIMA in capturing demand fluctuations in complex, high-variance domains such as logistics and workforce scheduling [16].

While LSTM offers high predictive power, it often lacks transparency. This trade-off between accuracy and interpretability is partially addressed by Prophet, an open-source model developed by Facebook. Prophet decomposes time series into trend, seasonality, and holiday effects [17, 18], offering strong accuracy with intuitive parameter control. It has shown robust performance in diverse

settings such as retail [19], energy demand [20], and financial forecasting [18], making it a compelling choice for firms balancing usability with precision.

To evaluate these models fairly, the study employs a triad of performance metrics: RMSE, which penalizes large forecast deviations; MAE, which captures average absolute error; and MAPE, which offers percentage-based interpretability [21]. Although MAPE can exaggerate errors when actual values are small, the combination of these metrics ensures a holistic assessment. Notably, comparative studies in call center staffing highlight how ARIMA and Prophet differ in their sensitivity to these metrics [22].

However, forecasting is only the first step. What follows is the critical challenge of resource allocation, which is turning predictions into optimal deployment strategies. To achieve this, the study drew from network flow optimization, a method that has gained traction in domains such as semiconductor supply chains [23], telecommunications [24], and workforce management [25]. These models treat staffing as a flow problem, balancing supply (employee availability), demand (workload forecasts), and operational constraints.

Solving such models efficiently requires a robust optimization engine. The study thus adopted CPLEX, a solver widely recognized for its speed and reliability in handling large-scale, resource-constrained scheduling problems [26–28]. CPLEX was integrated via Python to solve the formulated minimum-cost flow problem, ensuring alignment between forecasted demand and resource availability while honoring business constraints like work-hour caps, shared labor pools, and leave schedules [29, 30].

To ensure these technical outputs remain actionable for decision-makers, the final layer of the solution stack involved data visualization. Here, Power BI was selected for its cost-effectiveness, native Microsoft integration, and strength in rendering interactive dashboards [31, 32]. With built-in support for time-series plotting, matrix views, and slicers [33], it could empower stakeholders to explore forecast trends, track fulfillment levels, and simulate resourcing scenarios, bridging the gap between data science and operational strategy.

Despite these advancements, gaps in the literature persist. Most forecasting-optimization frameworks target product-based or static planning contexts, with limited relevance to the time-sensitive, high-variance nature of professional services. Few studies explored how interpretability and granularity trade-offs influence model selection in such environments. Moreover, visualization tools are rarely embedded into end-to-end systems, limiting their strategic impact.

This research addressed these gaps by developing a unified, dynamic, and interpretable pipeline that connected demand forecasting, resource optimization, and visualization, specifically tailored to the unique operational demands of professional service firms. In doing so, it not only enhanced model accuracy and usability but also ensured alignment with real-world decision-making needs.

3 METHODOLOGY

3.1 Data Understanding

To address the research questions, both quantitative and qualitative data were collected. Historical activities from the company's time-tracking system served as the primary dataset, complemented by operational insights from the team's manager. The dataset spans January 2, 2018 to December 24, 2024 and includes six key attributes. The predictive phase uses the first four: 1.) Date (DD/MM/YYYY), 2.) Billable (binary: 1 for client-related, revenue-generating tasks; 0 for internal activities), 3.) Duration (decimal) (task length, capped at 24 hours), and 4.) Client (binary: 1 or 2). The prescriptive model adds 5.) Number of Resources (six analysts excluding QA) and 6.) Working Hours Limit (8 hours/day, extendable to 10).

3.2 Data Analysis

3.2.1 Descriptive/Diagnostics Phase. This phase involved four key activities. First, summary statistics (mean, min, max, percentiles, standard deviation) were generated to compare demand variability between the two clients. Second, time series decomposition using seasonal_decompose from statsmodels was applied to extract trend and seasonality components. Third, the Augmented Dickey-Fuller (ADF) test assessed stationarity; a p-value ≤ 0.05 indicates suitability for ARIMA modeling. Lastly, autocorrelation analysis using ACF and PACF plots provided insight into lag dependencies, guiding selection of AR and MA terms in ARIMA.

3.2.2 Predictive Phase. The predictive phase was implemented in Google Colab using Python. Key attributes were selected, with the dataset filtered to billable entries and aggregated daily. To ensure continuity, the time series was resampled to daily frequency (freq='D') and missing values were addressed via linear interpolation. The cleaned dataset was then split into 60:40, 70:30, and 80:20 train-test ratios. Model tuning followed to identify optimal parameters for each forecasting approach.

Four forecasting models were implemented using Python. ARIMA, from the statsmodels.tsa.arima.model library, was tuned via grid search across combinations of (p, d, q) parameters, representing autoregressive lags, differencing for stationarity, and moving average terms, respectively. Holt-Winters Exponential Smoothing, via ExponentialSmoothing from statsmodels.tsa.holtwinters, was configured with various combinations of trend and seasonality types (additive or multiplicative), along with an optional damped trend component to model decaying trends [34, 35]. Prophet, imported from the prophet library, required reformatting the dataset into 'ds' (date) and 'y' (target). Key parameters included change_point_prior_scale for trend flexibility and seasonality_prior_scale for capturing recurring patterns, with higher values allowing greater adaptability [36]. LSTM, a deep learning model built using TensorFlow, involved data sequencing followed by a two-layer LSTM architecture with return_sequences, a dropout layer to mitigate overfitting, and a dense output layer. The model was compiled using the Adam optimizer and mean squared error loss to improve training accuracy [37].

3.2.3 Model Evaluation. Models were first tuned using optimal parameters identified through multiple train-test split validations. Final evaluation was then conducted on a 30-day holdout set using

RMSE, MAE, and MAPE to assess forecast accuracy. RMSE captured large errors, MAE reflected average deviation, and MAPE measured relative error, though it was less reliable for low-demand values. To address conflicting results, a weighted error score was computed as $(0.35 \times \text{MAE}) + (0.35 \times \text{RMSE}) + (0.30 \times \text{MAPE})$. The overall score also incorporated interpretability and training speed (0.25 each) to reflect practical deployment. Simpler and faster models received lower (better) scores, while complex or slower ones like LSTM scored higher. This ensured a consistent and fair comparison across accuracy, usability, and efficiency.

3.2.4 Prescriptive Phase. The model was translated as a Mixed-Integer Linear Programming (MILP) problem, designed to optimize the allocation of employees while meeting daily demand requirements. It accounted the following variables and parameters shown in Table 1. x_i , C_i , t represented hours availability of each employee, which could be allocated for a given client and day depending on needs. HC_{it} represented the demand nodes, or the service level requirements of each client for a given day. A concept of backlog was introduced through the E_{it-1} variable, which was the difference of hours supplied, and hours needed from the previous day. Hours required by each client and backlog hours were subject to non-negativity constraints, enforced through within=NonNegativeReals in the python code. In addition, overtime and employee availability attributes were represented by $o_{i,t}$ and $y_{i,t}$, respectively. All these were configured to the model via Var() function with iteration for each employee, client, and days.

The objective function aimed to minimize the excess resources in terms of hours availability, minimizing operational costs and improving utilization rate while satisfying demand requirements of each client. This formulation balanced resource utilization ensuring daily needs were covered by the team. Due to high confidentiality, it was modelled that regular cost was \$5 per hour, overtime cost was \$10 per hour, and penalty cost was \$20 per hour. Mathematically, it was expressed with the following formula in equation 1.

$$\sum_{i \in \{E1, E2, E3, E4, S1, S2\}, t} C_{\text{regular}} (x_i C_1 t + x_i C_2, t) + C_{\text{overtime}} (O_{i,t}) + C_{\text{penalty}} (E1_t + E2_t) \quad (1)$$

As shown in equations 2 to 4, this model accounted three constraints: (1) The first one was the demand satisfaction constraint, followed by (2) working hours limit, and (3) employee availability constraints. The first constraint represented the billable hours that need to be matched by employees. Next, the model put restriction on the maximum hours of work to ensure employee well-being. The third constraint was defined in the $y(i,t)$ expression, ensuring no work was assigned to an employee if they were out of office. This resulted to 0 capacity if the value was 1.

$$\sum_{i \in \{E1, E2, S1, S2\}} x_i C_1 t = HC_1 t + E1_{t-1} \quad (2)$$

$$\sum_{i \in \{E3, E4, S1, S2\}} x_i C_2 t = HC_2 t + E2_{t-1} \quad (3)$$

$$x_i C_1, t + x_i C_2, t \leq (8 + o_{i,t}) * (1 - y_{i,t}) \quad (4)$$

After formulating the objective, key variables, and operational constraints, the model proceeded to execution by calling the solver package in Pyomo. The study utilized the CPLEX solver, implemented in the code as solver = SolverFactory('cplex_direct'). Unlike traditional file-based approaches, where users must write or read

Table 1: Optimization Variables and Parameters

Variable	Definition	Thresholds
x_i, C_1, t	Hours supplied by employee i for client 1 on day t	$0 \leq x_i, C_1, t \leq 8 + o_{i,t}$
x_i, C_2, t	Hours supplied by employee i for client 2 on day t	$0 \leq x_i, C_1, t \leq 8 + o_{i,t}$
HC_1t	Forecasted hours required by client 1 on day t	Non-negative real numbers
HC_2t	Forecasted hours required by client 2 on day t	Non-negative real numbers
$E1_{t-1}$	Unmet hours needed by client 1 from the previous day (backlogs)	Non-negative real numbers
$E2_{t-1}$	Unmet hours needed by client 2 from the previous day (backlogs)	Non-negative real numbers
$o_{i,t}$	Overtime hours	$0 \leq o_{i,t} \leq 2$
$y_{i,t}$	Indicator whether an employee is on leave (1) or not (0)	Binary (0,1)
$C_{regular}$	Cost of regular working hours	\$5/hour (hypothetical)
$C_{overtime}$	Cost of overtime hours	\$10/hour (hypothetical)
$C_{penalty}$	Penalty cost of unmet demand/backlogs	\$20/hour (hypothetical)

data files [30], such as in MATLAB or Lingo, this plug-in allowed seamless interaction with IBM’s CPLEX solver through Python bindings [29]. Beyond efficiency, this choice was driven by its seamless integration and compatibility with the Python-based workflow, ensuring a streamlined optimization process.

3.3 Data Visualization

Custom visualizations were developed to enhance interpretability and support decision-making. A Sankey diagram was employed to depict the flow of billable hours from clients to employees, effectively illustrating workload distribution. Weekly bar charts, color-coded by client, were used to present trends and forecasted demand, while key summary metrics provided an overview of total weekly requirements. These visual components enabled stakeholders to quickly grasp demand dynamics and informed resource planning.

3.4 Ethical Considerations

This study utilized actual company operations data, requiring strict diligence in responsible data handling. Data privacy and confidentiality were observed by anonymizing the clients being catered in the study. No personal identifiable information was collected throughout the course of the study to adhere to data governance protocols of the company. The operations manager was informed on the extent of data usage to answer the study’s objectives. As part of data integrity and transparency process, the data were solely used for academic purposes only. Any processing and data manipulations performed were documented in the methodology section of the study.

3.5 Limitations

This study adopted a quantitative approach, which presented certain limitations. First, the analysis was restricted to output-based variables, such as billable hours, and was confined to group-level assessments; input-level factors like individual analyst skill, which could enhance the model, were unavailable. Second, the forecasting model utilized internal historical data and did not incorporate external economic indicators from the outsourcing country, which may influence demand variability. Third, caution was warranted when using the tool for long-range forecasting, as it did not consider rare but high-impact events (e.g., the COVID-19 pandemic). Lastly,

the effectiveness of the proposed system was highly dependent on consistent and accurate updating of time-tracking data.

4 RESULTS AND DISCUSSION

This section presented the findings of the time series analysis to develop a demand forecasting and resource allocation model for a market research firm. The results were organized to address the three main research questions: forecasting accuracy, optimal workforce distribution, and effective visualization.

4.1 Diagnostic Phase

The diagnostic results in Table 2 effectively mirrored the operational characteristics of each client. For instance, the upward trend in Client 1’s data was a clear indication of expanded syndicated research scope and consistent repeat contracts. Conversely, the volatile trend observed for Client 2 suggested the varied nature of their projects, which included periodic brand reports, ad-hoc custom research demand, or studies in new market domains or industries.

4.2 Predictive Phase

The study examined four models and were tested using two evaluation techniques: 1. multiple data splits for forecast stability checking, and 2. holdout evaluation for model selection. The first approach checked if models generalized well with varying amount of historical data used in training, whereas the second one simulated if model performed well in production setting, where future values were totally unknown.

4.2.1 Hyperparameter tuning using multiple data splits. Multiple train-test splits were used to evaluate model robustness and optimize hyperparameters, as shown in Figures 1, 2, 3 and 4. For ARIMA and Holt-Winters, larger training sizes consistently improved MAE and RMSE, particularly for Client 1, though MAPE remained volatile for Client 2 due to variable project types. Prophet showed sensitivity to changepoint prior scale, with better fit at higher flexibility but increased relative error at lower splits, especially for Client 2. LSTM maintained stable performance across all splits, showing less sensitivity to training size but strong absolute accuracy. These

Table 2: Diagnostic Results

Aspect	Client 1	Client 2
Trend Component	Moderately increasing trend	Moderately increasing trend
Seasonal Component	Weak	Weak
Residual Component	Random	Random
ACF Behavior	Slowly decays, suggesting MA component	Slowly decays, suggesting MA component
PACF Behavior	Significant lag at 1 and 2, suggesting AR component	Significant lag at 1 and 2, suggesting AR component
Stationarity	Stationary series	Stationary series
Model Type	Additive	Additive

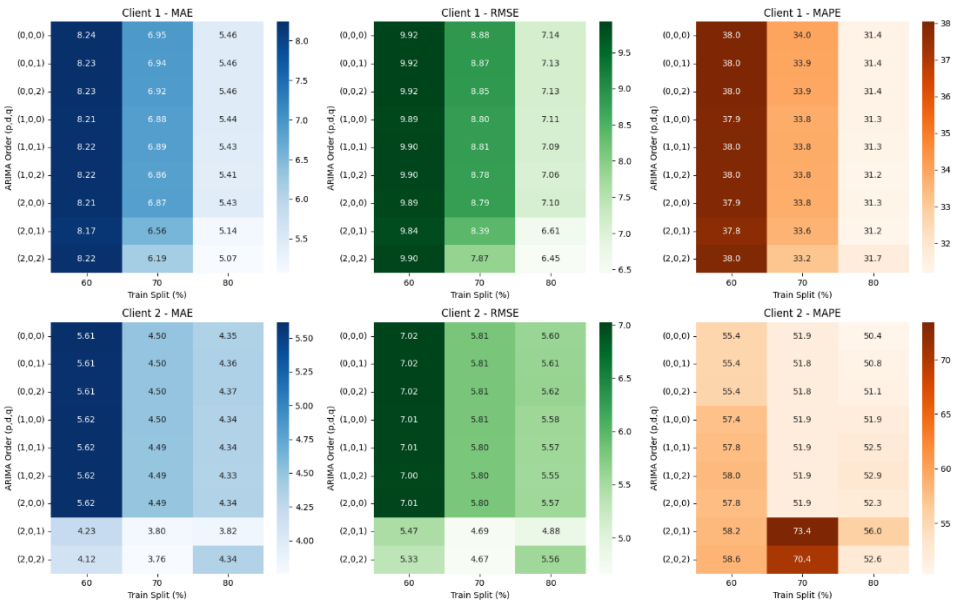


Figure 1: ARIMA Model Performance Metrics Across Settings and Training Splits. For Client 1, increasing training size improved model accuracy. Client 2 showed stronger absolute performance (lower MAE, RMSE), but higher MAPE, indicating larger relative error sensitivity.

results underscored the importance of split-based validation for selecting optimal configurations tailored to data volume and demand volatility.

4.2.2 Holdout Evaluation. A 30-day holdout evaluation confirmed that the Additive Prophet model outperformed all others, yielding the lowest weighted error for Client 1 (13.038) and showing strong results for Client 2 (27.786), as shown in Table 3. Prophet’s strength in handling trend shifts and event-triggered variations made it more adaptive to both stable and volatile demand patterns, outperforming ARIMA and Holt-Winters, particularly under noisy conditions. Although LSTM showed competitive accuracy, its higher complexity and data requirements reduced its practical viability. When interpretability and speed were factored in, Prophet achieved the lowest overall score (10.456), shown in Table 4, validating its suitability for operational forecasting in resource-constrained service environments.

4.3 Prescriptive Phase

Forecasted service demand was operationalized using a Pyomo-based optimization model, tested across three scenarios: baseline forecasts, positive RMSE-adjusted forecasts (to simulate under-forecasting), and negative RMSE-adjusted forecasts (to simulate over-forecasting). The model strictly respected capacity constraints, ensuring no employee exceeded the 10-hour daily cap and aligned resource allocations with forecasted demand across all dates, achieving near-zero gaps. As reflected in Figure 5, exclusivity rules were enforced, assigning fixed analysts to each client while shared resources (S1, S2) flexibly absorbed load variations. Utilization analysis confirmed efficient deployment, though slight overutilization (e.g., 106% for S2), shown in Table 5, in high-demand scenarios flagged areas for managerial attention. Cost analysis in Figure 6 revealed that small discrepancies in short-term forecasts could compound into significant cost gaps over time, reinforcing the value of dynamic, short-horizon planning. Weekly

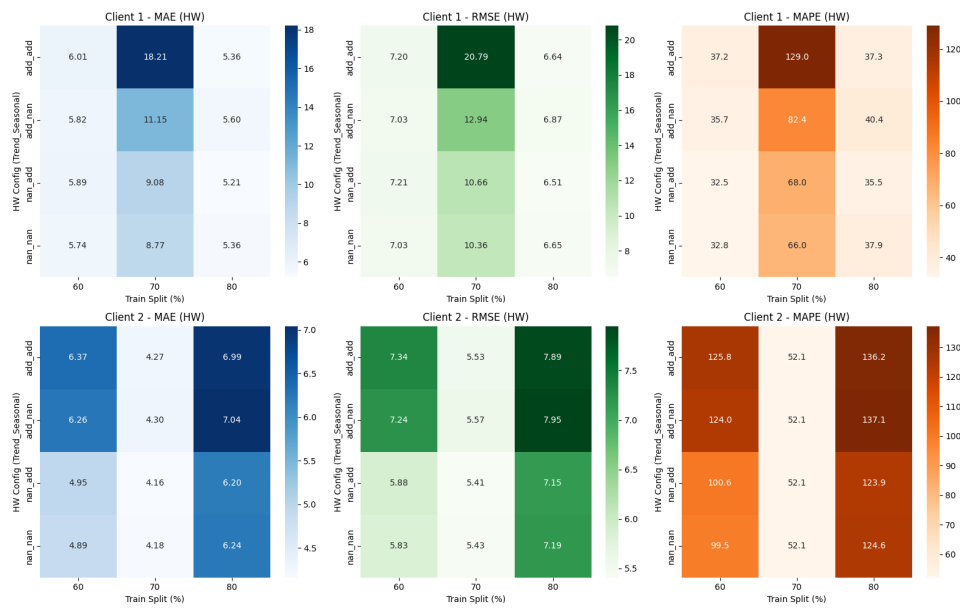


Figure 2: Holt-Winter's Model Performance Metrics Across Settings and Training Splits. For Client 1, the model achieved lowest errors at higher training sizes. In contrast, Client 2 displayed more volatile results with high MAPE values, especially under additive trend-seasonality settings, indicating sensitivity to seasonality assumptions and lower robustness compared to other models.

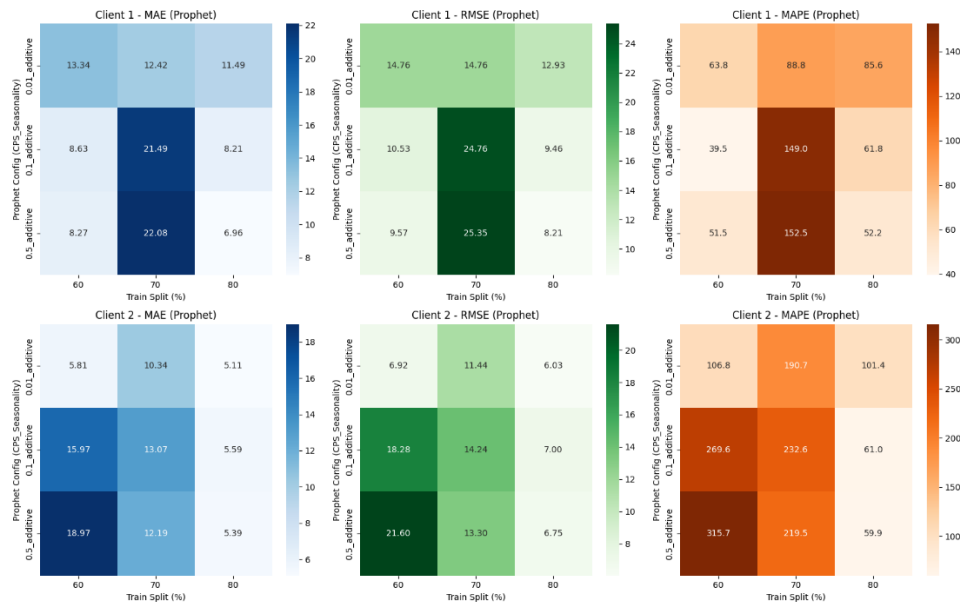


Figure 3: Prophet Model Performance Metrics Across Settings and Training Splits. Results showed variability across settings, particularly for Client 2 in lower train splits, indicating model sensitivity to parameter tuning.

or bi-weekly use of the tool was found optimal, as it allowed for cost-effective and responsive staffing aligned with shifting demand. This prescriptive framework not only optimized resource use under realistic operational constraints but also delivered actionable

insights for balancing service levels, workforce capacity, and labor costs in volatile service environments.

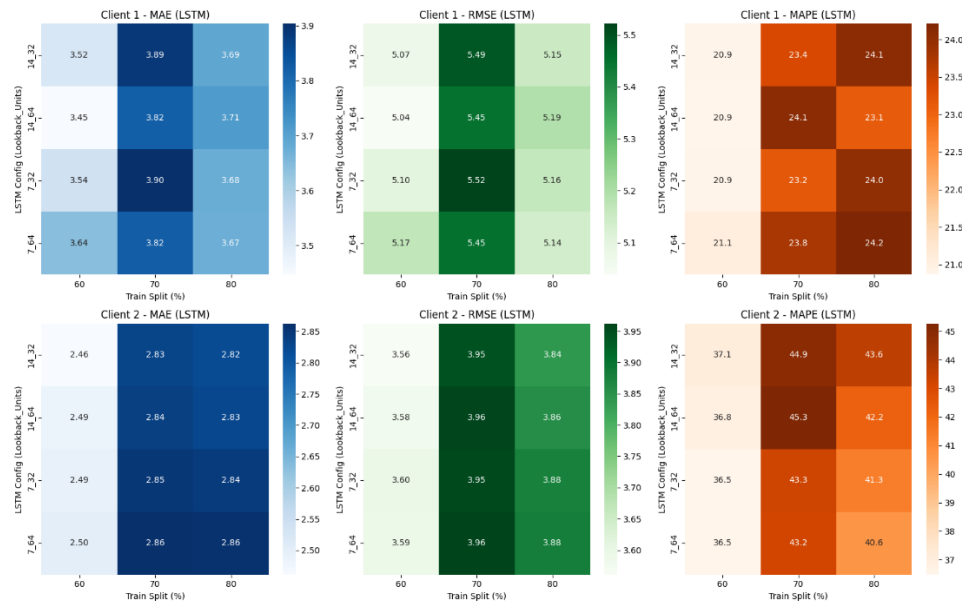


Figure 4: LSTM Model Performance Metrics Across Settings and Training Splits. Results showed stable performance across all splits and hyperparameters, with Client 1 showing slightly higher accuracy.

Table 3: Summary of Weighted Error Metrics

Model	Client	MAE (0.35)	RMSE (0.35)	MAPE (0.30)	Weighted Error Metrics
ARIMA (2,0,1)	1	5.970	7.528	54.799	21.164
Holt-Winter (None,None)	1	5.182	7.076	46.644	18.284
Additive Prophet (cps=0.1)	1	4.748	6.639	30.175	13.038
LSTM (lookback=7, units=32)	1	4.755	6.639	42.386	16.703
ARIMA (0,0,0)	2	4.253	4.785	71.669	24.664
Holt-Winter (None,Add)	2	3.335	4.535	102.363	33.463
Additive Prophet (cps=0.5)	2	3.317	4.057	84.017	27.786
LSTM (lookback=7, units=32)	2	3.170	4.036	84.369	27.833

Table 4: Overall Model Selection

Model	Average Error (0.50)	Interpretability (0.25)	Speed (0.25)	Weighted Score	Rank
ARIMA	22.914	0.250	0.250	11.582	2
Holt-Winter	25.874	0.250	0.250	13.062	4
Prophet	20.412	0.500	0.500	10.456	1
LSTM	22.268	1.000	1.000	11.634	3

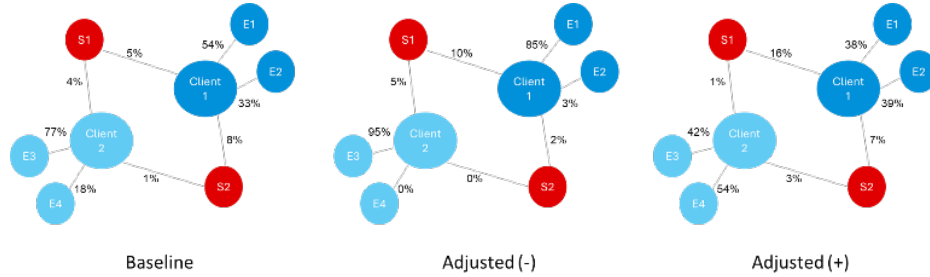
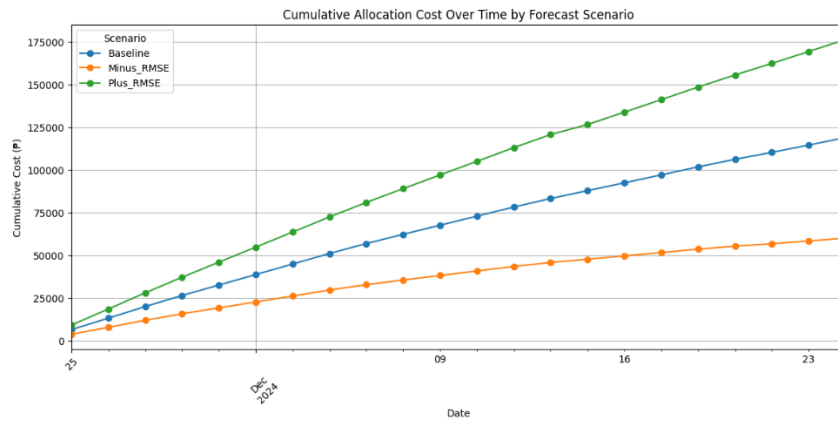
4.4 Visualization Phase

Forecasted demand and optimized resource allocations were visualized through an interactive Power BI dashboard, enabling real-time operational insights as shown in Figure 7. Key slicers allowed managers to filter by client, staff, scenario, and date range, supporting agile decision-making under uncertainty. Visual elements such as Sankey diagrams and bar charts highlighted workload distribution,

utilization, and free capacity, helping identify staffing imbalances. KPI cards monitored total demand, assigned hours, unmet demand, overtime cost, and service fulfillment, providing actionable indicators for performance and risk. Together, these tools facilitated dynamic, data-driven workforce planning aligned with fluctuating service requirements.

Table 5: Optimized Average Resource Utilization

Employees	Baseline	Adjusted (-)	Adjusted (+)
E1	89%	59%	97%
E2	55%	4%	96%
E3	100%	74%	75%
E4	25%	0%	100%
S1	88%	72%	42%
S2	71%	28%	106%

**Figure 5: Resource Distribution by Forecast Scenario****Figure 6: Cumulative Allocation Cost by Forecast Scenario**

5 CONCLUSION AND RECOMMENDATION

5.1 Conclusion

The Prophet model was seen as the most effective approach, balancing accuracy and operational usability, particularly in short-term planning contexts. Its performance improved with larger, regularly updated datasets, enabling dynamic, data-driven decisions. Incorporating prediction intervals into the optimization phase enhanced allocation precision by accounting for demand uncertainty. Optimization outputs respected key constraints such as working hours, employee availability, and resource assignments (dedicated vs. shared). Integrated within an interactive Power BI dashboard, the combined forecasting-optimization system provided actionable insights through visual tools like Sankey diagrams and KPI panels, supporting agile responses in volatile service environments.

5.2 Recommendation

Given the critical role of training size in forecasting accuracy, this study recommended establishing a robust data warehousing framework to ensure the availability, continuity, and temporal integrity of historical data, particularly during system transitions. For future research, researchers may incorporate adaptive learning mechanisms and real-time analysis to support continuous improvement of both forecasting and optimization processes in dynamic service environments. Furthermore, expanding the model to include input-based variables (e.g., skill levels, project complexity) and integrating external economic indicators could further enhance predictive accuracy.



Figure 7: Resource Mapping and Utilization Dashboard

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References

[1] Ugboke, L. N., & Ehugbo, I. (2021). Effect of resource utilization techniques on competitive advantage of selected firms in Nigerian telecommunications industry. *International Journal of Academic Multidisciplinary Research (IJAMR)*, 5(11), 106-117. Retrieved from https://www.researchgate.net/publication/359504374_Effect_of_Resource_Utilization_Techniques_on_Competitive_Advantage_of_Selected_Firms_in_Nigerian_Telecommunications_Industry

[2] Sharma, S., & Singh, S. (2022). RLS model for forecasting resource fulfilment in IT and consulting industry. *Metamorphosis: A Journal of Management Research*, 21(1), 1–10. <https://doi.org/10.1177/09726225221075086>

[3] Boadi-Sarpong, A. B., Agyei, P. A., Mensah, E., & Boateng, P. A. (2023). Resource allocation strategies for maintaining competitiveness and achieving innovation success. *International Journal of Research and Scientific Innovation (IJRSI)*, 10(7). <https://doi.org/10.51244/IJRSI.2023.10724>

[4] Gartner. (n.d.). NetSuite planning and budgeting alternatives & competitors. *Gartner Peer Insights*. Retrieved from <https://www.gartner.com/reviews/market/financial-planning-software/vendor/oracle/product/netsuite-planning-and-budgeting/alternatives>

[5] AltexSoft. (2022, March 1). Demand forecasting methods: Using machine learning to see the future of sales. *AltexSoft*. Retrieved from <https://www.altexsoft.com/blog/demand-forecasting-methods-using-machine-learning/>

[6] Business Money. (2022, July 25). Demand forecasting: What happens when it fails. *Business Money*. Retrieved from <https://www.business-money.com/announcements/demand-forecasting-what-happens-when-it-fails/>

[7] Sutanto, E. M. (2000). Forecasting: The key to successful human resource management. *Jurnal Manajemen & Kewirausahaan*, 2(1), 1–8. Retrieved from <https://www.researchgate.net/publication/50434617>

[8] Jose, J. (2022). Introduction to time series analysis and its applications. *Christ University*. Retrieved from https://www.researchgate.net/publication/362389180_INTRODUCTION_TO_TIME_SERIES_ANALYSIS_AND_ITS_APPLICATIONS

[9] Ghosh, S. (2020). Forecasting of demand using ARIMA model. *American Journal of Applied Mathematics and Computing*, 1(2), 13–18. <https://ajamc.smartsociety.org/wp-content/uploads/2020/09/paper4.pdf>

[10] Girotra, R. (2022). Workforce planning using time series (Master’s dissertation). *Trinity College Dublin*. <https://publications.scss.tcd.ie/theses/diss/2022/TCD-SCSS-DISSERTATION-2022-070.pdf>

[11] Devi, K. L., & Valli, S. (2023). Time series-based workload prediction using the statistical hybrid model for the cloud environment. *Computing*, 105(2), 353–374. <https://doi.org/10.1007/s00607-022-01129-7>

[12] Kumar, L., Khedlekar, S., & Sharma, R. (2024). A comparative assessment of Holt-Winters exponential smoothing and autoregressive integrated moving average for inventory optimization in supply chains. *Supply Chain Analytics*, 8, Article 100084. <https://doi.org/10.1016/j.sca.2024.100084>

- [13] Wang, C.-C., Chien, C.-H., & Trappey, A. J. C. (2021). On the application of ARIMA and LSTM to predict order demand based on short lead time and on-time delivery requirements. *Processes*, 9(7), 1157. <https://doi.org/10.3390/pr9071157>
- [14] Swapnarekha, H., Behera, H. S., Nayak, J., Naik, B., & Kumar, P. S. (2021). Multiplicative Holt's Winter model for trend analysis and forecasting of COVID-19 spread in India. *SN Computer Science*, 2, Article 416. <https://doi.org/10.1007/s42979-021-00808-0>
- [15] Yadav, M., Patil, R., Bhirud, B., & Chavle, P. L. (2023). LSTM model-based demand forecasting in supply chain using efficient optimizer. Retrieved from https://www.researchgate.net/publication/373555323_LSTM_Model_based_Demand_Forecasting_in_Supply_Chain_using_Efficient_Optimizer
- [16] Terrada, L., El Khaili, M., & Ouajji, H. (2022). Demand forecasting model using deep learning methods for supply chain management 4.0. *International Journal of Advanced Computer Science and Applications*, 13(5), 704–711. <https://doi.org/10.14569/IJACSA.2022.0130581>
- [17] Facebook Research. (2017, February). Prophet: Forecasting at scale. Facebook Research. Retrieved from <https://research.facebook.com/blog/2017/2/prophet-forecasting-at-scale/>
- [18] Sharma, K., Bhalla, R., & Ganesan, G. (2022). Time series forecasting using FB-Prophet. In R. Zwigelaar, G. Ganesan, Q. S. Cheng, K.-L. Du, & S. R. Satti (Eds.), *Proceedings of the Algorithms, Computing and Mathematics Conference 2022 (ACM 2022)* (pp. 59–64). CEUR Workshop Proceedings, Vol-3445. https://ceur-ws.org/Vol-3445/PAPER_07.pdf
- [19] Negre, P., Alonso, R. S., Prieto, J., García, Ó., & de-la-Fuente-Valentín, L. (2024). Prediction of footwear demand using Prophet and SARIMA. *Expert Systems with Applications*, 234, 124512. <https://reunir.unir.net/handle/123456789/17524>
- [20] Long, C., Yu, C., & Li, T. (2022). Prophet-based medium and long-term electricity load forecasting research. *Journal of Physics: Conference Series*, 2356, 012002. <https://doi.org/10.1088/1742-6596/2356/1/012002>
- [21] Jierula, A., Wang, S., OH, T.-M., & Wang, P. (2021). Study on accuracy metrics for evaluating the predictions of damage locations in deep piles using artificial neural networks with acoustic emission data. *Applied Sciences*, 11(5), 2314. <https://doi.org/10.3390/app11052314>
- [22] Seder, A., & Alnasser, R. (2023). Workforce management for Salik call center (Master's thesis). Rochester Institute of Technology. Retrieved from [https://scholarworks.rit.edu/cgi/viewcontent.cgi?article=\\$12635&context=\\$theses](https://scholarworks.rit.edu/cgi/viewcontent.cgi?article=$12635&context=$theses)
- [23] Wang, Y., Zhang, H., Yuan, C., Li, X., & Jiang, Z. (2024). A network flow approach to optimal scheduling in supply chain logistics. *arXiv*. <https://doi.org/10.48550/arXiv.2411.17544>
- [24] Wang, C. C. (2022). Network flow optimization using reinforcement learning (Doctoral dissertation). Aalborg University. https://vbn.aau.dk/ws/portalfiles/portal/549541544/PHD_CCW.pdf
- [25] Ercsey, Z., & Kovács, Z. (2023). Multicommodity network flow model of a human resource allocation problem considering time periods. *Central European Journal of Operations Research*. <https://doi.org/10.1007/s10100-023-00868-y>
- [26] Mohammadi, S., PourKarimi, L., & Reinert, K. (2023). A mathematical programming approach for resource allocation of data analysis workflows on heterogeneous clusters. *The Journal of Supercomputing*, 79(17), 19019–19048. <https://doi.org/10.1007/s11227-023-05325-w>
- [27] Fang, Z., Wu, Q., & Guan, D. (2022). Towards optimal resources allocation in cloud manufacturing: New task decomposition strategy and service composition model. *Security and Communication Networks*, 2022, Article 5019584. <https://doi.org/10.1155/2022/5019584>
- [28] Kia, R., Shahnazari-Shahrezaei, P., & Zabihi, S. (2016). Solving a multi-objective mathematical model for a multi-skilled project scheduling problem by CPLEX solver. *Proceedings of the 2016 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, 1220–1224. <https://doi.org/10.1109/IEEM.2016.7798072>
- [29] Pyomo Developers. (n.d.). Pyomo CPLEX direct solver module. Pyomo Documentation. Retrieved March 18 2025, from https://pyomo.readthedocs.io/en/6.8.1/_modules/pyomo/solvers/plugins/solvers/cplex_direct.html
- [30] OR Stack Exchange. (n.d.). Difference between CPLEX direct and CPLEX in Pyomo SolverFactory. OR Stack Exchange. Retrieved March 18 2025, from <https://or.stackexchange.com/questions/9412/difference-between-cplex-direct-and-cplex-in-pyomo-solverfactory>
- [31] Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. (2023). A comprehensive review of data visualization tools: Features, strengths, and weaknesses. *International Journal of Computer Engineering in Research Trends*, 10(1), 10–20. <https://doi.org/10.22362/ijcert/2023/v10/i01/v10i0102>
- [32] Tirupati, K. K., Joshi, A., Singh, S. P., Chhapola, A., Jain, S., & Gupta, A. (2024). Leveraging Power BI for enhanced data visualization and business intelligence. *Universal Research Reports*, 10(2), 676–711. <https://doi.org/10.36676/urr.v10.i2.1375>
- [33] Microsoft. (n.d.). Power BI visualization types for reports and Q&A. Microsoft Learn. Retrieved March 2025, from <https://learn.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-types-for-reports-and-q-and-a>
- [34] Pleños, M. C. F. (2022). Time series forecasting using Holt-Winters exponential smoothing: Application to abaca fiber data. *Problems of World Agriculture*, 22(2), 17–29. <https://doi.org/10.22630/PRS.2022.22.2.6>
- [35] StatsModels. (n.d.). statsmodels.tsa.holtwinters.ExponentialSmoothing. StatsModels Documentation. Retrieved February 26, 2025, from <https://www.statsmodels.org/dev/generated/statsmodels.tsa.holtwinters.ExponentialSmoothing.html>
- [36] Facebook. (n.d.). Trend changepoints in Prophet. Facebook Prophet Documentation. Retrieved February 26, 2025, from https://facebook.github.io/prophet/docs/trend_changepoints.html
- [37] Arya, M., & Sastry, G. H. (2023). Effective LSTM neural network with Adam optimizer for improving frost prediction in agriculture data stream. In A. E. Hassanien, M. A. Elhoseny, & A. T. Azar (Eds.), *Modelling and development of intelligent systems* (pp. 3–17). Springer. https://doi.org/10.1007/978-3-031-27034-5_1